

Test 34: DIRECT AB-cloud vs ? (full loaded set)

Verdict: FAIL | Generated: 2026-09-15 07:51:46 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Statistical: unfolded-spacing KS (effect size $D < 0.10$) + Montgomery $R?(s)$ vs ?-zeros and GUE. full 50k range, not ?-5000.

METHOD: Statistical: unfolded-spacing KS (effect size $D < 0.10$) + Montgomery $R?(s)$ vs ?-zeros and GUE. full 50k range, not ?-5000.

Test 34 HARDCORE pass 2: $n=5$ realizations, $R? ds=0.05/s_max=4.0$, per-realization KS stability.

Test 34 HARDCORE: $D=0.1096$ ($p=2.236e-186$), per-real D med 0.1153 (spread 0.0923..0.1189), $|?|?R?|?=0.0002$, $d_GUE=0.8764$.

Test 34b [HARDCORE pass 2]: 4 sub-checks, 1 failed ? WARN

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 34 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[07:30:25.651] [+27816.680s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[07:34:35.595] [+28066.623s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[07:38:52.443] [+28316.621s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

FULL COMPUTATION LOG

[07:43:11.074] [+28582.102s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[07:47:23.751] [+28834.779s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

??

TEST 34 (pass 2, HARDCORE): direct AB vs ? ? deep validation

CONSOLE CAPTURE

Test 34 HARDCORE pass 2: n=5 AB realizations at 96x96 + ? unfolding (50000 zeros) ? ? 12?17 min

??

? RUNTIME: $n=5$ AB realizations at 96x96 + ? unfolding (50000 zeros) ? ? 12?17 min

Config: $N_v=2$, $q=1.00$, $?=0.5000$, $W=1.0$ (base protocol)

real 1/5: 5528 spacings, $D_k(?) = 0.0923$

real 2/5: 5527 spacings, $D_k(?) = 0.1188$

real 3/5: 5528 spacings, $D_k(?) = 0.1153$

real 4/5: 5528 spacings, D_k(?) = 0.1065

real 5/5: 5528 spacings, D_k(?) = 0.1189

Pooled: 27639 AB spacings vs 49999 ? spacings | KS D = 0.1096, p = 2.236e-186

Per-realization D(?): min 0.0923 / median 0.1153 / max 0.1189 (n=5)

?|R?_AB ? R?_?|? (ds=0.05, s?4) = 0.0002 (target < 0.10)

d_GUE = 0.8764 vs d_Poisson = 0.9998 ? closer to GUE

??

KS composite WARN ? p = 2.236e-186, D = 0.1096 (p>0.01 or D<0.10)

?|?R?|? < 0.10 (fine) PASS ? ds=0.05, s_max=4.0: 0.0002

closer to GUE PASS ? d_GUE=0.8764 < d_Pois=0.9998

KS ensemble stability PASS ? D spread 0.0923..0.1189 (median 0.1153)

Test 34b [HARDCORE pass 2]: 4 sub-checks, 1 failed ? WARN